



Tree Intelligence Assistant

1. Introduction

The **Tree Intelligence Assistant** is a Python-based AI application developed using **Streamlit** that helps users identify and recommend tree species based on environmental conditions and basic inputs. The project is designed for students and nature enthusiasts to understand how artificial intelligence can be applied to environmental sustainability and biodiversity awareness.

This project demonstrates the use of rule-based logic and machine learning concepts through an interactive web interface.

2. Objectives

- To recommend suitable tree species based on soil type, temperature, and rainfall.
 - To provide a simple, user-friendly interface for tree recommendation.
 - To demonstrate the use of Python, Streamlit, and basic AI logic.
 - To build an educational and practical environmental application.
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3. Technologies Used

- **Programming Language:** Python 3.10
 - **Frontend Framework:** Streamlit
 - **Libraries:**
 - NumPy
 - Pillow (PIL)
 - TensorFlow (optional for image model)
 - **Development Environment:** Visual Studio Code
 - **Operating System:** Windows 10
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4. System Architecture

The application follows a simple architecture:

1. **User Interface (Streamlit):**
2. Collects user input (soil type, temperature, rainfall)
3. Displays recommended trees

4. **Business Logic Layer:**

5. Processes user inputs

6. Applies rule-based recommendation logic

7. **Output Layer:**

8. Displays tree recommendations dynamically

5. Functional Modules

5.1 Recommend Trees by Location

This module recommends tree species based on:

- Soil Type (Sandy, Clay, Loam)
- Average Temperature
- Annual Rainfall

The system uses predefined environmental rules to suggest trees that grow well in given conditions.

5.2 User Interaction Module

- Dropdowns and sliders for inputs
 - Button-based execution
 - Real-time results displayed on the browser
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6. Implementation Details

6.1 Input Parameters

- **Soil Type:** Selected using dropdown
- **Temperature:** Selected using slider (0–50 °C)
- **Rainfall:** Selected using slider (0–4000 mm)

6.2 Recommendation Logic

- Sandy soil → Coconut, Casuarina
- Clay soil → Teak, Banyan
- Loam soil → Neem, Peepal
- High rainfall → Peepal
- Moderate temperature → Neem

Duplicate results are removed before displaying.

7. User Interface Design

- Clean dark-themed layout
- Sidebar-based navigation
- Clear labels and instructions
- Responsive and browser-based UI

8. Output Screens

The application displays:

- Recommended tree species list
- Icons for better visual clarity
- Success messages after recommendation

9. Advantages

- Easy to use
- No external database required
- Fast execution
- Beginner-friendly AI project
- Useful for educational purposes

10. Limitations

- Rule-based recommendations (not fully data-driven)
- Limited tree species
- Image classification module optional

11. Future Enhancements

- Add real machine learning models
 - Integrate GPS-based location input
 - Expand tree database
 - Add map visualization
 - Deploy on cloud (Streamlit Cloud)
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12. Conclusion

The **Tree Intelligence Assistant** successfully demonstrates how artificial intelligence concepts can be applied to environmental sustainability. The project provides an interactive platform to learn tree selection based on environmental factors and serves as a strong foundation for future AI-based ecological applications.

13. References

- Streamlit Documentation
 - Python Official Documentation
 - Environmental Tree Plantation Guides
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